# Bootstrap Confidence Intervals

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## Introduction

The bootstrap offers several ways to construct confidence intervals from resamples. They differ in accuracy (first- vs second-order correctness), in assumptions, and in complexity. Choosing well among them is part of responsible bootstrap use.

## Prerequisites

Bootstrap, confidence intervals.

## Theory

For a statistic $\hat{\theta}$ with bootstrap replicates $\hat{\theta}^{*(1)}, \ldots, \hat{\theta}^{*(B)}$:

- **Percentile**: $[\hat{\theta}^*_{(\alpha/2)}, \hat{\theta}^*_{(1-\alpha/2)}]$. Simple and intuitive; assumes symmetry of the bootstrap distribution.
- **Basic (pivotal)**: $[2\hat{\theta} - \hat{\theta}^*_{(1-\alpha/2)}, 2\hat{\theta} - \hat{\theta}^*_{(\alpha/2)}]$. Uses the bootstrap distribution of $\hat{\theta}^* - \hat{\theta}$ as a pivot.
- **Studentised (bootstrap-t)**: requires bootstrap SE, generally most accurate but computationally heavier.
- **BCa (bias-corrected and accelerated)**: corrects for bias and skewness; the default modern choice.

**Coverage**: percentile is first-order accurate (error $O(n^{-1/2})$); BCa and studentised are second-order ($O(n^{-1})$).

## Assumptions

Same as bootstrap generally; specific methods may assume additional smoothness.

## R Implementation

```r
library(boot)
set.seed(2026)

x <- rlnorm(40, meanlog = 0, sdlog = 0.8)   # skewed data

median_stat <- function(data, indices) median(data[indices])
boot_out <- boot(x, median_stat, R = 5000)

ci <- boot.ci(boot_out, type = c("norm", "basic", "perc", "bca"))
ci

# Studentised bootstrap
median_with_var <- function(data, indices) {
  d <- data[indices]
  md <- median(d)
  # Jackknife variance of median
  jk <- sapply(seq_along(d), function(i) median(d[-i]))
  c(md, var(jk) * (length(d) - 1))
```

```
}
boot_t <- boot(x, median_with_var, R = 5000)
boot.ci(boot_t, type = "stud")
```

### Output & Results

```
BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS

Normal        (0.728, 1.241)
Basic         (0.712, 1.221)
Percentile    (0.735, 1.244)
BCa           (0.800, 1.297)

Studentized  (0.729, 1.341)
```

For skewed data, BCa shifts the interval to the right, correcting for the bias in the percentile CI.

### Interpretation

"The median was 0.97 with a 95 % BCa bootstrap CI of (0.80, 1.30) based on 5000 resamples."

### Practical Tips

- Default to BCa CIs unless computational constraints matter; they correct for bias and skew at modest additional cost.
- For very small $B$ (e.g., $B < 1000$), percentile and BCa can be unstable at the tails; increase $B$.
- Studentised (boot-t) CIs are the most accurate but require a variance estimate of the statistic (jackknife or formula); worth it for critical inference.
- Normal bootstrap CIs (using mean +/- 1.96 SE) are the crudest; avoid them unless the bootstrap distribution is nearly normal.
- Report which CI method was used and the number of bootstrap resamples.

### For Reviewers

**What to look for in a paper using this method.**

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

### See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests